

COURSE SYLLABUS

AI Essentials

COURSE NUMBER

AIE-001

DURATION

1 day · 5 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This comprehensive one-day course introduces participants to fundamental AI capabilities in modern work environments. Students will master prompt engineering techniques, image generation, document processing, and spreadsheet automation while learning critical information security, safety protocols, and legal compliance requirements. The platform-neutral curriculum uses multiple AI tools to provide practical, hands-on experience in responsible AI implementation.

AT COURSE COMPLETION

After completing this course, you will be able to:

- Master prompt engineering techniques to effectively communicate with AI systems and achieve desired outcomes
 - Generate and manipulate images using AI tools for practical business applications across marketing and communications
 - Automate document processing and data extraction workflows to reduce manual effort and improve accuracy
 - Implement spreadsheet automation to enhance productivity through AI-powered data transformation and analysis
 - Apply information security, safety protocols, and legal compliance requirements throughout AI implementation initiatives
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PREREQUISITES

- Basic computer literacy and web browser navigation
 - Familiarity with common office productivity tools
 - Understanding of basic business workflows and processes
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COURSE OUTLINE

MODULE 1 Prompt Engineering Fundamentals

90 min

Explore core techniques for crafting effective prompts that guide AI systems to produce accurate, relevant, and useful outputs. Learn strategies for context setting, role assignment, and iterative refinement.

LESSONS

- Lesson 1:** Introduction to Prompt Engineering
- Lesson 2:** Anatomy of Effective Prompts
- Lesson 3:** Role Assignment and Persona-Based Prompting
- Lesson 4:** Chain-of-Thought and Step-by-Step Reasoning
- Lesson 5:** Few-Shot Learning and Example-Based Guidance
- Lesson 6:** Temperature and Parameter Adjustments
- Lesson 7:** Common Pitfalls and Iterative Refinement

LAB EXERCISES

- Building and Refining Effective AI Prompts

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Construct clear and specific prompts using context, instructions, and constraints
- Apply role-based prompting techniques to improve response quality and relevance
- Implement iterative refinement strategies to optimize AI outputs
- Evaluate prompt effectiveness using measurable criteria and testing frameworks

Master techniques for creating, modifying, and refining images using AI tools. Understand parameters that control style, composition, and quality while learning practical applications for business communications and marketing.

LESSONS

- Lesson 1:** Introduction to AI Image Generation
- Lesson 2:** Crafting Effective Image Prompts
- Lesson 3:** Composition and Visual Control Techniques
- Lesson 4:** Image Editing and Refinement Techniques
- Lesson 5:** Quality Control and Platform Selection
- Lesson 6:** Copyright, Ethics, and Commercial Usage

LAB EXERCISES

- Creating Marketing Visuals with DALL-E 3

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Generate custom images from text descriptions using style parameters and composition guidelines
- Modify existing images through inpainting, outpainting, and variation techniques
- Apply quality control criteria to evaluate AI-generated visual content
- Implement image generation workflows for specific business use cases

MODULE 3 Automated Document Processing and Data Extraction 85 min

Learn to leverage AI for document understanding, content extraction, and workflow automation. Develop skills in processing unstructured content and transforming it into actionable structured data.

LESSONS

- Lesson 1:** Introduction to AI-Powered Document Processing
- Lesson 2:** Document Understanding Technologies
- Lesson 3:** Entity Recognition and Data Extraction
- Lesson 4:** Document Classification and Routing
- Lesson 5:** Data Validation and Quality Assurance
- Lesson 6:** Integration and End-to-End Automation

LAB EXERCISES

- Building an Invoice Processing Pipeline with Document AI

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Extract structured data from unstructured documents using AI parsing techniques
- Implement document classification and routing workflows based on content analysis
- Validate and clean extracted data to ensure accuracy and completeness
- Design end-to-end document processing pipelines for business automation

MODULE 4 Spreadsheet Automation and Data Intelligence

80 min

Harness AI capabilities to automate spreadsheet tasks, generate complex formulas, and transform raw data into actionable insights. Learn techniques for data cleaning, enrichment, and advanced analysis.

LESSONS

- Lesson 1:** Introduction to AI-Powered Spreadsheet Automation
- Lesson 2:** AI-Assisted Formula Generation and Debugging
- Lesson 3:** Data Cleaning and Standardization
- Lesson 4:** Pattern Recognition and Anomaly Detection
- Lesson 5:** Data Enrichment and External Lookup Automation
- Lesson 6:** Predictive Analytics and Automated Categorization

LAB EXERCISES

- Hands-On Lab: Automating Spreadsheet Workflows

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Automate data cleaning and normalization tasks using AI-powered tools
- Generate complex formulas and calculations through natural language instructions
- Implement data enrichment workflows that augment existing datasets with external information
- Create automated reporting and visualization pipelines for business intelligence

MODULE 5 Information Security, Safety, and Legal Compliance

80 min

Understand critical security considerations, safety protocols, and legal requirements for responsible AI implementation. Learn to identify risks, apply mitigation strategies, and ensure compliance with regulations.

LESSONS

- Lesson 1:** The Critical Importance of AI Security and Compliance
- Lesson 2:** Data Privacy and Preventing Information Disclosure
- Lesson 3:** Access Control and Authentication for AI Systems
- Lesson 4:** Bias Detection and Fairness in AI Systems
- Lesson 5:** Harmful Content Prevention and Output Filtering
- Lesson 6:** Data Protection Regulations: GDPR, CCPA, and Beyond
- Lesson 7:** Industry-Specific Regulations and Compliance Requirements
- Lesson 8:** Intellectual Property and Copyright in AI Context
- Lesson 9:** Incident Response and Audit Trail Maintenance

LAB EXERCISES

- Evaluating AI Safety and Compliance Controls

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Identify information security risks in AI workflows including data leakage and unauthorized access
- Apply safety protocols to prevent harmful outputs and ensure responsible AI usage
- Evaluate AI systems for bias, fairness, and ethical considerations
- Implement compliance measures for data protection regulations and industry standards

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4, 5

SKILLS TO BE INTEGRATED

- Craft multi-step prompts with context, role assignment, and output formatting to guide AI responses toward specific business objectives
- Apply iterative refinement techniques to improve prompt effectiveness through systematic testing and adjustment
- Generate custom images from text descriptions by specifying style parameters, composition elements, and quality settings
- Edit and modify AI-generated images using inpainting, outpainting, and variation techniques to meet specific visual requirements
- Extract structured data from unstructured documents using AI parsing tools and validation workflows
- Automate document classification and routing based on content analysis and business rules
- Create automated data transformation pipelines that clean, normalize, and enrich spreadsheet information
- Implement formula generation and complex calculations using natural language instructions to AI assistants
- Evaluate AI outputs for potential security risks including data leakage, unauthorized access patterns, and compliance violations
- Apply responsible AI principles by assessing bias, fairness, transparency, and accountability in automated decision-making systems

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.